Lecture 16: Finite Elements in 1D

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120
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After this lecture you should be able to

- Derive the weak form of $-(c\,u')' = f$ by testing against v and integrating by parts once.
- Say why a natural boundary condition needs nothing imposed and an essential one must be built into the functions.
- Show that the transpose of d/dx is $-d/dx$ from the same integration by parts.
- Assemble $\boldsymbol{KU} = \boldsymbol{F}$ from hat functions, one element at a time, for any $c(x)$.
- Say what quadratic and cubic elements add, and how the error falls with h for each.
- Set up the seiche in a parabolic lake as the problem $\boldsymbol{KU} = \lambda \boldsymbol{MU}$.

1 From a spring chain to a rod

In Lecture 12 a chain of springs gave $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}$ in three steps: elongation, Hooke's law, force balance. Now let the chain become a rod. Take N short pieces of length $h = 1/N$, with displacement u_j at the point $x_j = jh$. Let $c_{j+1/2}$ be the stiffness of the material between x_j and x_{j+1}. That piece is a spring of constant $c_{j+1/2}/h$: a shorter piece of the same material is harder to stretch. Let f_j be the load per unit length at x_j. Per unit length, the three steps are

$$\begin{aligned} e_{j+1/2} &= \frac{u_{j+1} - u_j}{h} &\longrightarrow \quad e(x) &= \frac{du}{dx}, \\ w_{j+1/2} &= c_{j+1/2}\, e_{j+1/2} &\longrightarrow \quad w(x) &= c(x)\, e(x), \\ -\frac{w_{j+1/2} - w_{j-1/2}}{h} &= f_j &\longrightarrow \quad -\frac{dw}{dx} &= f(x). \end{aligned}$$

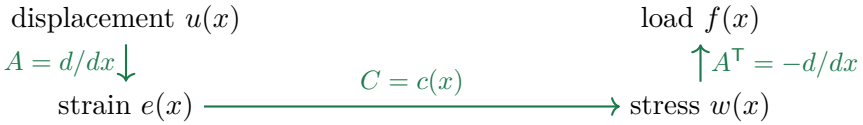
Here e is the strain, w the stress (the tension in the rod) and f the load per unit length. The last line is the force balance at x_j, which reads $w_{j+1/2} - w_{j-1/2} + hf_j = 0$, divided by h. Together the three limits give our model problem,

$$-\frac{d}{dx}\left(c(x)\,\frac{du}{dx}\right) = f(x), \qquad 0 \le x \le 1, \tag{1}$$

with **free–fixed** boundary conditions,

$$w(0) = c(0)\,u'(0) = 0, \qquad u(1) = 0. \tag{2}$$

The rod need not be uniform, so the stiffness is a function $c(x) > 0$. At the free end $x = 0$ there is no tension, and the end $x = 1$ is pinned.



The matrix $\boldsymbol{A}$ has become d/dx, the matrix $\boldsymbol{C}$ has become multiplication by $c(x)$, and the last step suggests that $\boldsymbol{A}^{\mathsf{T}}$ has become $-d/dx$. Section 3 shows that this is correct, so (1) is the continuous form of $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}\,\boldsymbol{u} = \boldsymbol{f}$.

Finite differences replace the derivatives in (1) by differences. The finite element method starts from another statement of the same problem, the weak form, and discretises that.

2 The weak form

The idea is to test an equality of functions with integrals. If $f_1(x) = f_2(x)$ on $[0,1]$ then

$$\int_0^1 v(x) f_1(x)\,dx = \int_0^1 v(x) f_2(x)\,dx$$

for every function $v(x)$. The converse also holds. Taking $v(x) = \delta(x-a)$ gives $f_1(a) = f_2(a)$, and a can be any point. The function v is a **test function**. So "weak" does not mean approximate: the integrals for all v say the same thing as $f_1 = f_2$. The approximation comes later, when we keep only finitely many v.

2.1 Testing the differential equation

Write $w = c\,u'$ for the stress. Multiply (1) by a test function v and integrate:

$$-\int_0^1 v\,\frac{dw}{dx}\,dx = \int_0^1 v\,f\,dx. \tag{3}$$

The left side contains two derivatives of u. Move one of them onto v with the product rule $(vw)' = v'w + vw'$. Integrating that rule over $[0,1]$ gives

$$\Big[vw\Big]_0^1 = \int_0^1 v'w\,dx + \int_0^1 vw'\,dx, \qquad \text{so} \qquad -\int_0^1 v\,w'\,dx = \int_0^1 v'w\,dx - \Big[vw\Big]_0^1.$$

Put this into (3) and replace w by $c\,u'$:

$$\int_0^1 c\,u'\,v'\,dx \;-\; \Big(c(1)u'(1)\,v(1) - c(0)u'(0)\,v(0)\Big) = \int_0^1 v\,f\,dx. \tag{4}$$

The two boundary terms go away for two different reasons.

- At $x = 0$ the boundary condition gives $c(0)u'(0) = 0$, so the term vanishes whatever $v(0)$ is. There is no restriction on v. This is a **natural** boundary condition.
- At $x = 1$ we do not know $u'(1)$: the pin supplies whatever force is needed. The term vanishes only if we require $v(1) = 0$. So the condition $u(1) = 0$ is imposed on the functions themselves, u and v alike. This is an **essential** boundary condition, and a test function with $v(1) = 0$ is called **admissible**.

The weak form

Find u with $u(1) = 0$ such that

$$\int_0^1 c(x)\,u'(x)\,v'(x)\,dx \;=\; \int_0^1 f(x)\,v(x)\,dx \qquad \text{for every } v \text{ with } v(1) = 0. \tag{5}$$

Each of u and v now carries one derivative. Functions with kinks are allowed, because a piecewise linear function has a piecewise constant first derivative and this integral makes sense for it. A rod with a jump in c needs no special treatment either.

Example 2.1 (Checking the weak form by hand)**. Given.** $c = 1$ and $f = 6x$, free at 0, fixed at 1. The solution is $u = 1 - x^3$, with $u'(0) = 0$ and $u(1) = 0$. Take the admissible test function $v = 1 - x$.

Find. (a) What is the left side of (5)? (b) What is the right side?

Answer. (a) With $u' = -3x^2$ and $v' = -1$,

$$\int_0^1 u'v'\,dx = \int_0^1 3x^2\,dx = 1.$$

(b) $\int_0^1 fv\,dx = \int_0^1 (6x - 6x^2)\,dx = 3 - 2 = 1$. The two sides agree. A test function that is not admissible fails: see Section S.1 (reading).

2.2 Why "natural"

Equation (5) does not mention $u'(0) = 0$, yet the condition follows from it. Suppose u is smooth and satisfies (5). Integrate the left side by parts back again, using $v(1) = 0$:

$$\int_0^1 c\,u'\,v'\,dx = \Big[c\,u'\,v\Big]_0^1 - \int_0^1 (c\,u')'\,v\,dx = -c(0)u'(0)\,v(0) - \int_0^1 (c\,u')'\,v\,dx.$$

Then (5) becomes

$$\int_0^1 \big[-(c\,u')' - f\big]\,v\,dx \;-\; c(0)u'(0)\,v(0) = 0 \qquad \text{for every } v \text{ with } v(1) = 0. \tag{6}$$

First take test functions that also have $v(0) = 0$. The integral vanishes for all of them, so $-(c\,u')' = f$. The integral is therefore zero for every v, and (6) leaves $c(0)u'(0)\,v(0) = 0$. Choosing $v(0) = 1$ gives $c(0)u'(0) = 0$. The weak form contains the differential equation and the natural condition. It does not contain the essential condition, which has to be built in.

If both ends were fixed, both boundary terms would need $v(0) = v(1) = 0$. A physical reading of v as a virtual displacement is in Section S.2 (reading).

3 The transpose of d/dx

For a matrix, $\boldsymbol{A}^{\mathsf{T}}$ is the matrix with $\langle \boldsymbol{x}, \boldsymbol{Ay}\rangle = \left\langle \boldsymbol{A}^{\mathsf{T}}\boldsymbol{x}, \boldsymbol{y}\right\rangle$ for all $\boldsymbol{x}$ and $\boldsymbol{y}$, because $\boldsymbol{x}^{\mathsf{T}}(\boldsymbol{Ay}) = (\boldsymbol{A}^{\mathsf{T}}\boldsymbol{x})^{\mathsf{T}}\boldsymbol{y}$. This statement also makes sense for operators on functions. For functions on $[0, 1]$ the inner product is $\langle p, q\rangle = \int_0^1 p(x)\,q(x)\,dx$. Take $A = d/dx$ and integrate by parts as in Section 2:

$$\langle w, Au\rangle = \int_0^1 w\,u'\,dx = \int_0^1 (-w')\,u\,dx + \Big[w\,u\Big]_0^1 = \langle -w', u\rangle + w(1)u(1) - w(0)u(0).$$

The boundary conditions (2) remove both end terms: $u(1) = 0$ and $w(0) = 0$. Then $\langle w, Au\rangle = \langle -w', u\rangle$, which says

$$A^{\mathsf{T}} = -\frac{d}{dx}.$$

The conditions belong to the two spaces: A acts on displacements with $u(1) = 0$, and A^{T} acts on stresses with $w(0) = 0$. Other boundary conditions give other spaces, as they gave $\boldsymbol{K}$, $\boldsymbol{T}$, $\boldsymbol{B}$ and $\boldsymbol{C}$ in Lecture 2. The same result from a difference matrix is in Section S.3 (reading).

Now look at the left side of the weak form. Moving A^{T} across gives

$$\left\langle v, A^{\mathsf{T}}CA\,u\right\rangle = \langle Av, CA\,u\rangle = \int_0^1 c\,v'\,u'\,dx,$$

so (5) is $\langle Av, CAu\rangle = \langle v, f\rangle$, the function version of $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}^{\mathsf{T}}\boldsymbol{C}\boldsymbol{A}\,\boldsymbol{u} = \boldsymbol{v}^{\mathsf{T}}\boldsymbol{f}$. Integrating by parts is the step that turns A^{T} acting on CAu into A acting on the test function. Exchanging u and v leaves the left side unchanged, so the matrix we build next will be symmetric.

4 Trial functions, test functions and hats

From the weak form (5) we want a linear system $\boldsymbol{KU} = \boldsymbol{F}$. Two restrictions produce it.

1. Instead of every v, use N **test functions**. This gives N equations.
2. Instead of every u, use combinations $U(x) = \sum_j U_j\,\phi_j(x)$ of N **trial functions** ϕ_j. This gives N unknown numbers U_j.

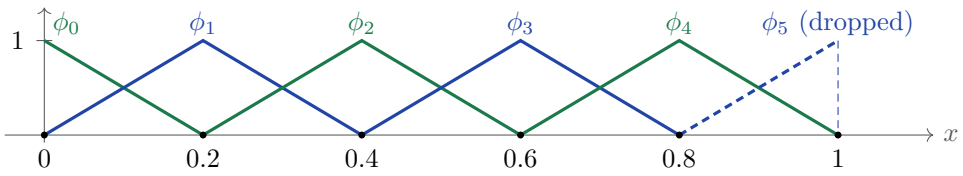
The system is square: one equation per test function, one unknown per trial function. Choosing the test functions equal to the trial functions is **Galerkin's** choice, and we use it from now on. The method is called *finite elements* when each ϕ_j is zero outside a few small pieces of the interval, the elements.

4.1 Hat functions

Divide $[0,1]$ into N elements of width $h = 1/N$, with nodes $x_j = jh$. The hat function ϕ_j is the piecewise linear function that equals 1 at node j and 0 at every other node:

$$\phi_j(x) = \begin{cases} (x - x_{j-1})/h & \text{on } [x_{j-1}, x_j],\ \text{slope } +1/h, \\ (x_{j+1} - x)/h & \text{on } [x_j, x_{j+1}],\ \text{slope } -1/h, \\ 0 & \text{elsewhere.} \end{cases}$$

At the ends of the interval only half a hat is left. For $N = 5$, $h = 0.2$:



Since $\phi_j(x_i)$ is 1 for $i = j$ and 0 otherwise, the coefficient U_j is the value of $U(x)$ at node j, and $U(x)$ joins the nodal values by straight lines.

Which hats stay

The fixed end $u(1) = 0$ is essential, so ϕ_N is dropped: there is no unknown at $x = 1$, and no test function there. The free end is natural, so the half hat ϕ_0 stays, and U_0 is an unknown. On N elements the free–fixed rod has the N unknowns $U_0, \ldots, U_{N-1}$.

4.2 Assembling $\boldsymbol{KU} = \boldsymbol{F}$

Put $U(x) = \sum_{j=0}^{N-1} U_j \phi_j(x)$ into (5) with $v = \phi_i$, and pull the sum and the numbers U_j out of the integral:

$$\int_0^1 c(x) \left(\sum_j U_j \, \phi_j'(x) \right) \phi_i'(x) \, dx = \int_0^1 f(x) \, \phi_i(x) \, dx,$$
$$\sum_j \left(\int_0^1 c(x) \, \phi_i'(x) \, \phi_j'(x) \, dx \right) U_j = \int_0^1 f(x) \, \phi_i(x) \, dx.$$

For $i = 0, \ldots, N-1$ this is the system $\boldsymbol{KU} = \boldsymbol{F}$ with

$$K_{ij} = \int_0^1 c(x) \, \phi_i'(x) \, \phi_j'(x) \, dx, \qquad F_i = \int_0^1 f(x) \, \phi_i(x) \, dx. \tag{7}$$

By inspection $K_{ij} = K_{ji}$.

Finite support makes K sparse

If $|i - j| > 1$ then ϕ_i and ϕ_j have no element in common, the integrand in (7) is zero everywhere, and $K_{ij} = 0$. With hats, $\boldsymbol{K}$ is tridiagonal for every $c(x)$. Sparsity comes from the small supports of the trial functions.

Compute $\boldsymbol{K}$ one element at a time. On element k, the interval $[x_{k-1}, x_k]$, only two hats are non-zero: ϕ_{k-1} with slope $-1/h$ and ϕ_k with slope $+1/h$. Their slopes are constants, so they come out of the integral. Writing $\bar{c}_k = \frac{1}{h} \int_{x_{k-1}}^{x_k} c \, dx$ for the mean of c on the element, so that $\int_{x_{k-1}}^{x_k} c \, dx = h \, \bar{c}_k$, the three integrals are

$$\int_{x_{k-1}}^{x_k} c \, (\phi_{k-1}')^2 \, dx = \Big(-\frac{1}{h}\Big)\Big(-\frac{1}{h}\Big) \, h \, \bar{c}_k = \frac{\bar{c}_k}{h},$$
$$\int_{x_{k-1}}^{x_k} c \, (\phi_k')^2 \, dx = \Big(+\frac{1}{h}\Big)\Big(+\frac{1}{h}\Big) \, h \, \bar{c}_k = \frac{\bar{c}_k}{h},$$
$$\int_{x_{k-1}}^{x_k} c \, \phi_{k-1}' \, \phi_k' \, dx = \Big(-\frac{1}{h}\Big)\Big(+\frac{1}{h}\Big) \, h \, \bar{c}_k = -\frac{\bar{c}_k}{h}.$$

The element matrix for (ϕ_{k-1}, ϕ_k) is therefore

$$\frac{\bar{c}_k}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Assembly means adding each element matrix into rows and columns $k-1$, k of $\boldsymbol{K}$. For element N only the entry for ϕ_{N-1} is kept, because ϕ_N was dropped. The result is the spring chain of Lecture 12: each element acts as a spring with constant $\bar{c}_k/h$, and $\boldsymbol{K} = \boldsymbol{A}^{\mathsf{T}} \boldsymbol{C} \boldsymbol{A}$, with $\boldsymbol{A}$ the matrix of differences $u_k - u_{k-1}$ and $\boldsymbol{C} = \mathrm{diag}(\bar{c}_k/h)$.

The right side is assembled the same way. Element k adds the integral of $f\,\phi_{k-1}$ over the element to F_{k-1} and the integral of $f\,\phi_k$ to F_k. For constant f each of these is $\frac{1}{2}fh$. A point load $f = \delta(x-a)$ gives $F_i = \int_0^1 \delta(x-a)\,\phi_i\,dx = \phi_i(a)$, so only the hats that are non-zero at a appear. On a non-uniform grid with element widths h_k, the same steps give the element matrix $(\bar{c}_k/h_k)\left[\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right]$.

For $c = 1$ every element matrix is $\frac{1}{h}\left[\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right]$. An interior hat has two elements, so $K_{jj} = 2/h$. Neighbouring hats share one element, so $K_{j,j\pm1} = -1/h$. The half hat ϕ_0 has one element, so $K_{00} = 1/h$. On the grid above,

$$\boldsymbol{K} = \frac{1}{h}\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix} = \frac{1}{h}\boldsymbol{T}.$$

This is the free–fixed matrix $\boldsymbol{T}$ of Lecture 2, where finite differences produced it. Dividing by h gives $\frac{1}{h^2}\boldsymbol{TU} = \boldsymbol{F}/h$, the form of Lecture 2. Here the 1 in the corner comes from the half hat at the free end, with no separate argument about that end. For varying c, for a non-uniform grid or for other elements the two methods generally give different matrices.

5 A worked example: the stepped rod

Example 5.1 (Stepped rod)**. Given.** A rod made of three pieces of equal length, $h = \frac{1}{3}$, with $c = 3, 2, 1$ from left to right. The load is $f = 1$, free at 0, fixed at 1. Use hats, one element per piece.

Find. (a) What is $\boldsymbol{K}$? (b) What is $\boldsymbol{F}$? (c) What are U_0, U_1, U_2? (d) How do they compare with the exact solution at the nodes?

Answer. (a) The element constants are $c/h = 9, 6, 3$. Add the three element matrices into rows and columns $(0,1)$, then $(1,2)$, then row and column 2 alone, since ϕ_3 is dropped:

$$\boldsymbol{K} = \begin{bmatrix} 9 & -9 & 0 \\ -9 & 9+6 & -6 \\ 0 & -6 & 6+3 \end{bmatrix} = \begin{bmatrix} 9 & -9 & 0 \\ -9 & 15 & -6 \\ 0 & -6 & 9 \end{bmatrix}.$$

(b) $F_i = \int_0^1 \phi_i\,dx$ is the area under the hat. The half hat has area $\frac{1}{2}h = \frac{1}{6}$ and the full hats have area $h = \frac{1}{3}$, so $\boldsymbol{F} = (\frac{1}{6}, \frac{1}{3}, \frac{1}{3})$.

(c) Add row 1 to row 2, and then the new row 2 to row 3:

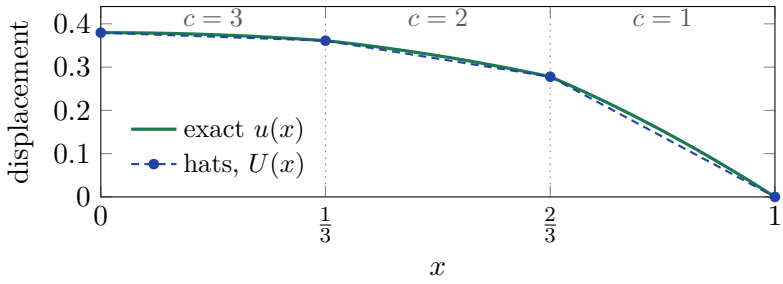
$$9U_0 - 9U_1 = \tfrac{1}{6}, \qquad 6U_1 - 6U_2 = \tfrac{1}{6} + \tfrac{1}{3} = \tfrac{1}{2}, \qquad 3U_2 = \tfrac{1}{2} + \tfrac{1}{3} = \tfrac{5}{6}.$$

Back substitution gives $U_2 = \frac{5}{18}$, then $U_1 = \frac{5}{18} + \frac{1}{12} = \frac{13}{36}$, then $U_0 = \frac{13}{36} + \frac{1}{54} = \frac{41}{108}$.

(d) From $-w' = 1$ and $w(0) = 0$ we get $w = -x$, so $u' = -x/c$ on each piece. Integrate back from $u(1) = 0$:

$$u(\tfrac{2}{3}) = \int_{2/3}^{1} \frac{x}{1}\,dx = \frac{5}{18}, \qquad u(\tfrac{1}{3}) = \frac{5}{18} + \int_{1/3}^{2/3} \frac{x}{2}\,dx = \frac{13}{36}, \qquad u(0) = \frac{13}{36} + \int_{0}^{1/3} \frac{x}{3}\,dx = \frac{41}{108}.$$

These are the same three numbers: the nodal values are exact. Section S.4 shows why this holds whenever c is constant on each element. The slope is not exact. At the free end $U'(0) = 3(U_1 - U_0) = -\frac{1}{18}$, while $u'(0) = 0$. The natural condition is reached only as $h \to 0$.



The elimination in (c) has a physical reading. Each new row is the force balance on the part of the rod to the left of a spring. The right sides $\frac{1}{6}, \frac{1}{2}, \frac{5}{6}$ are the sizes of the three spring forces. The rod is in compression: the stresses are $w = -\frac{1}{6}, -\frac{1}{2}, -\frac{5}{6}$, the values of $w = -x$ at the mid-elements. Lecture 18 uses this rod again.

6 Quadratic and cubic elements

Galerkin's method works with any trial space, and hats are the simplest choice. To do better, enlarge the space: add functions and keep the same weak form and the same formula (7). This section gives the two ideas and the result. The details are in Sections S.5 and S.6 (reading).

6.1 Quadratics: hats and bubbles

Between two nodes the true u bends away from the chord joining its end values. For a parabola the gap is exact. Take the element $[0, h]$ with u'' constant, and let $\text{gap}(x)$ be u minus its chord. It vanishes at $x = 0$ and $x = h$, and its second derivative is u'' because the chord has none. The quadratic with these properties is

$$\text{gap}(x) = -\tfrac{1}{2}\,u''\,x(h-x), \qquad \text{so at mid-element} \qquad \text{gap}(\tfrac{h}{2}) = -\tfrac{1}{2}\,u'' \cdot \tfrac{h}{2} \cdot \tfrac{h}{2} = -\frac{h^2}{8}\,u''. \tag{8}$$

The gap at mid-element is exactly $\frac{h^2}{8}\,|u''|$. In the figure of Example 5.1, the third element has $h = \frac{1}{3}$ and $u'' = -1$, and the gap is $\frac{1}{72}$. For a general smooth u this is the leading term, with u'' taken at mid-element.

Write $x = sh$ in (8). Then $x(h-x) = h^2 s(1-s)$ and $\text{gap} = -\frac{h^2}{8}\,u'' \cdot 4s(1-s)$: the gap is a multiple of one function. That function is the **bubble**. On element k, with $s = (x - x_{k-1})/h$,

$$b_k(x) = 4s(1-s),$$

and $b_k = 0$ outside the element. It equals 1 at mid-element and vanishes at both nodes, so adding bubbles curves $U(x)$ inside the elements and leaves the meaning of the U_j unchanged: $U(x) = \sum_j U_j\,\phi_j(x) + \sum_k B_k\,b_k(x)$.

The Taylor series gives the new order. Linear functions match 1 and x, and the first term they miss is $\frac{1}{2}u''x^2$. With a quadratic on each element we also match x^2. The first term missed is $\frac{1}{6}u'''x^3$, so the error should fall to $\mathcal{O}(h^3)$. When c is constant on each element, hats and bubbles decouple: the nodal values do not change, and the gain is between the nodes.

6.2 Cubics: a value and a slope at each node

For the rod a jump in slope does no harm. For a beam it does. A beam of length L and bending stiffness EI under load f obeys $(EI\,u'')'' = f$. Testing against v and integrating by parts twice

gives

$$\int_0^L EI\, u''\, v''\, dx = \int_0^L f\, v\, dx.$$

This weak form contains second derivatives. A kink in u puts a delta function into u'', and $\int (u'')^2\, dx$ would be infinite. So the trial functions need continuous slope: slopes must match at the nodes. The unknowns at each node are then a value and a slope. A cubic $a_0 + a_1 x + a_2 x^2 + a_3 x^3$ has four free parameters on each element, which is what is needed: two conditions at each end. A cubic also matches x^3. The first term missed is $\frac{1}{24} u'''' x^4$, so the error should fall to $\mathcal{O}(h^4)$.

Example 6.1 (Accuracy of the three elements)**. Given.** The rod with $c = 1 + x$, free at 0, fixed at 1, under the load

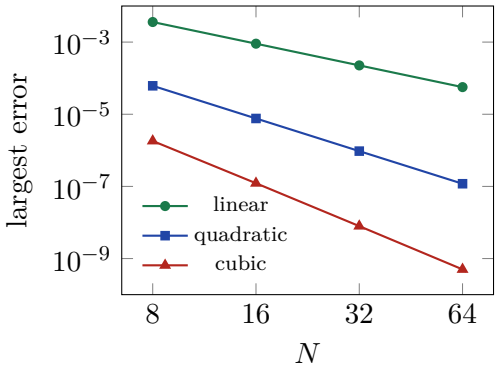
$$f = \frac{\pi}{2} \sin \frac{\pi x}{2} + (1 + x)\Big(\frac{\pi}{2}\Big)^2 \cos \frac{\pi x}{2},$$

whose exact solution is $u = \cos(\pi x/2)$ (Section S.7, reading). On $N = 8, 16, 32, 64$ equal elements, use hats, hats with bubbles, and the cubics of Section S.6. Take the largest of $|U - u|$ over 21 sample points in each element.

Find. (a) By what factor does the error fall when h is halved? (b) How do the three compare with about 16 unknowns?

Answer. (a) The table lists the errors. They fall by factors of 4, 8 and about 16, which are the orders h^2, h^3 and h^4.

N	linear	quadratic	cubic
8	$3.60 \cdot 10^{-3}$	$6.14 \cdot 10^{-5}$	$1.82 \cdot 10^{-6}$
16	$9.03 \cdot 10^{-4}$	$7.64 \cdot 10^{-6}$	$1.22 \cdot 10^{-7}$
32	$2.26 \cdot 10^{-4}$	$9.51 \cdot 10^{-7}$	$7.93 \cdot 10^{-9}$
64	$5.65 \cdot 10^{-5}$	$1.19 \cdot 10^{-7}$	$5.01 \cdot 10^{-10}$
ratios on halving h:			
$8 \to 16$	3.99	8.04	14.9
$16 \to 32$	4.00	8.03	15.4
$32 \to 64$	4.00	8.02	15.8



(b) With N elements the unknowns number N (linear), $2N$ (quadratic) and $2N + 1$ (cubic). So linear with $N = 16$, quadratic with $N = 8$ and cubic with $N = 8$ use 16, 16 and 17 unknowns, and their errors are $9.03 \cdot 10^{-4}$, $6.14 \cdot 10^{-5}$ and $1.82 \cdot 10^{-6}$. Here c varies inside the elements, so the nodal values of the hats are not exact: for $N = 8$ the largest error at a node is $1.54 \cdot 10^{-3}$.

7 A seiche in a parabolic lake

A seiche is a standing wave in a closed basin. In this section $h(x)$ is the water depth, not an element width, and g is the acceleration of gravity. For long surface gravity waves the height η of the surface and the horizontal velocity q obey $\eta_t + (h\, q)_x = 0$ (mass) and $q_t + g\, \eta_x = 0$ (momentum). Differentiate the first equation in t and put in the second: $\eta_{tt} = -(h\, q_t)_x = (g\, h\, \eta_x)_x$. A standing wave $\eta(x) \cos \omega t$ has $\eta_{tt} = -\omega^2 \eta \cos \omega t$, so

$$-\big(g\, h(x)\, \eta'\big)' = \omega^2\, \eta, \qquad -a \le x \le a, \qquad h(x) = h_0\Big(1 - \frac{x^2}{a^2}\Big). \tag{9}$$

This is the rod with $c(x) = g\,h(x)$, for a long lake of constant width whose depth varies along its length as a parabola, with depth h_0 in the middle and depth zero at the two ends $x = \pm a$. The weak form gives the condition at the shore. Multiply by v and integrate by parts as in Section 2:

$$\int_{-a}^{a} g\,h\,\eta'\,v'\,dx - \Big[g\,h\,\eta'\,v\Big]_{-a}^{a} = \omega^2 \int_{-a}^{a} \eta\,v\,dx.$$

Since $h(\pm a) = 0$, the boundary term vanishes for every v as long as η' stays bounded at the shore. By the momentum equation $h\,\eta'$ is proportional to the flux $h\,q$, so this says that no water crosses the shore. This is the natural boundary condition: nothing has to be imposed on v or on η. Every hat stays, including the two half hats at the shores.

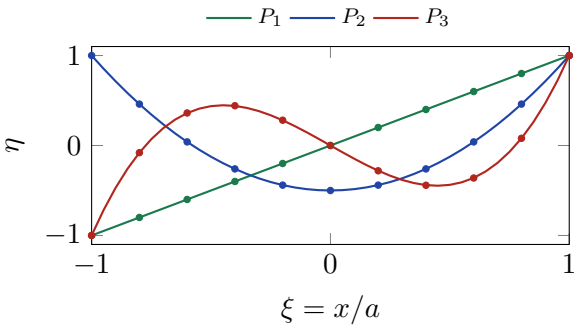
The exact modes are known. Put $x = a\xi$, so that $d/dx = \frac{1}{a}\,d/d\xi$ and $h = h_0(1-\xi^2)$:

$$-\frac{g\,h_0}{a^2}\Big((1-\xi^2)\,\eta_\xi\Big)_\xi = \omega^2\,\eta, \qquad \text{that is} \qquad -\Big((1-\xi^2)\,\eta_\xi\Big)_\xi = \lambda\,\eta, \qquad \lambda = \frac{\omega^2 a^2}{g\,h_0}.$$

This is Legendre's equation. Its bounded solutions are the Legendre polynomials $P_n(\xi)$, with $\lambda = n(n+1)$. For example $P_1 = \xi$ gives $-(1-\xi^2)' = 2\xi$, so $\lambda = 2$. Next $P_2 = \frac{1}{2}(3\xi^2 - 1)$ has $\eta_\xi = 3\xi$, so $-(3\xi - 3\xi^3)_\xi = 9\xi^2 - 3 = 6\,P_2$ and $\lambda = 6$. The third curve in the figure below is $P_3 = \frac{1}{2}(5\xi^3 - 3\xi)$, with $\lambda = 12$.

For the finite element version use hats on $[-1, 1]$, with $K_{ij} = \int_{-1}^{1}(1-\xi^2)\,\phi_i'\,\phi_j'\,d\xi$. Testing the right side gives $\lambda \sum_j (\int \phi_i \phi_j\,d\xi) U_j$, so the unknown function itself brings in the **mass matrix** $M_{ij} = \int_{-1}^{1} \phi_i\,\phi_j\,d\xi$, and the problem is $\boldsymbol{KU} = \lambda \boldsymbol{MU}$, a **generalised eigenvalue problem**. It is the problem behind $\boldsymbol{M\ddot{u} + Ku = 0}$ in Lecture 13, where the masses sat on the diagonal of $\boldsymbol{M}$. Here $\boldsymbol{M}$ is tridiagonal, because neighbouring hats overlap. Both matrices are symmetric and $\boldsymbol{M}$ is positive definite, so the λ are real. In Python, `scipy.linalg.eigh(K, M)` solves it. With 80 elements there are 81 unknowns:

n	exact $n(n+1)$	hats, 80 elements
0	0	0.000
1	2	2.000
2	6	6.002
3	12	12.011
4	20	20.042



The dots are every eighth computed nodal value, scaled to 1 at the shore. The value $n = 0$ is the constant mode, the mean water level, which does not move. The value $n = 1$ is exact because $P_1 = \xi$ is a straight line, which lies in the trial space.

Example 7.1 (Period of the first seiche)**. Given.** The lake above with $a = 5$ km, $h_0 = 20$ m, and $g = 9.81$ m/s^2.

Find. (a) What is ω for $n = 1$? (b) What is the period?

Answer. (a) From $\omega^2 a^2/(g\,h_0) = n(n+1) = 2$,

$$\omega = \frac{\sqrt{2\,g\,h_0}}{a} = \frac{\sqrt{2 \cdot 9.81 \cdot 20}}{5000} = \frac{19.81}{5000} = 3.96 \times 10^{-3} \text{ rad/s}.$$

(b) $T = 2\pi/\omega = 1586$ s, about 26 minutes.

Where this is going

The next lecture leaves one dimension. On a square grid the second difference becomes the five-point stencil, and $\boldsymbol{K}$ becomes the block matrix $\boldsymbol{K}_{2D}$. Lecture 18 then brings the weak form and the assembly of this lecture to two dimensions.

Reading

Strang, *Computational Science and Engineering*, §3.1 (Differential Equations of Equilibrium) and §3.6 (The Finite Element Method); §3.2 (Cubic Splines and Fourth Order Equations) for beams and piecewise cubics.

Appendix: finite elements in Python

The script below assembles $\boldsymbol{K}$ and $\boldsymbol{F}$ from hats, solves the stepped rod of Example 5.1, prints the linear column and the nodal error of Example 6.1, and finds the seiche eigenvalues. The other two columns of that table come from the same loop with three functions on each element (two hats and a bubble) or four (two values and two slopes). It needs NumPy and SciPy.

```
"""Lecture 16: hat functions in 1D (NumPy and SciPy)."""
import numpy as np
from scipy.linalg import eigh

g, w = np.polynomial.legendre.leggauss(8)    # Gauss rule on [-1, 1]
S, W = (g + 1) / 2, w / 2                    # the same on [0, 1]
ke = np.array([[1, -1], [-1, 1]])            # element matrix
me = np.array([[2, 1], [1, 2]])              # element mass matrix

def solve(c, f, N):
    """Hats on [0, 1], free at 0, fixed at 1: returns U_0, ..., U_N."""
    h = 1 / N
    K, F = np.zeros((N + 1, N + 1)), np.zeros(N + 1)
    for e in range(N):
        x = (e + S) * h                      # Gauss points of element e
        i = [e, e + 1]
        K[np.ix_(i, i)] += np.sum(W * c(x)) / h * ke
        F[i] += h * np.array([1 - S, S]) @ (W * f(x))
    U = np.zeros(N + 1)
    U[:N] = np.linalg.solve(K[:N, :N], F[:N])    # no unknown for u(1)
    return U

# 1. the stepped rod
c_step = lambda x: np.select([x < 1 / 3, x < 2 / 3], [3.0, 2.0], 1.0)
U = solve(c_step, lambda x: 1 + 0 * x, 3)
print("stepped rod, 108 U:", np.round(108 * U[:3], 6))

# 2. the hats in the accuracy example
exact = lambda x: np.cos(np.pi * x / 2)
c = lambda x: 1 + x
f = lambda x: (np.pi / 2 * np.sin(np.pi * x / 2)
               + (1 + x) * (np.pi / 2)**2 * np.cos(np.pi * x / 2))
print("   N    linear")
```

```
for N in (8, 16, 32, 64):
    x = np.linspace(0, 1, 20 * N + 1)        # 21 points in each element
    U = np.interp(x, np.arange(N + 1) / N, solve(c, f, N))
    print(f"{N:4d}  {np.abs(U - exact(x)).max():.2e}")
nodal = np.abs(solve(c, f, 8) - exact(np.arange(9) / 8)).max()
print(f"linear, N = 8, largest nodal error: {nodal:.2e}")

# 3. seiche: hats on [-1, 1], every node kept
N, d = 80, 2 / 80
K, M = np.zeros((N + 1, N + 1)), np.zeros((N + 1, N + 1))
for e in range(N):
    x = -1 + (e + S) * d
    i = [e, e + 1]
    K[np.ix_(i, i)] += np.sum(W * (1 - x**2)) / d * ke
    M[np.ix_(i, i)] += d / 6 * me
lam = eigh(K, M, eigvals_only=True)[:5]
print("seiche, 80 elements:", " ".join(f"{abs(v):.3f}" for v in lam))
T = 2 * np.pi * 5000 / np.sqrt(2 * 9.81 * 20)
print(f"n = 1 period: {T:.0f} s = {T / 60:.1f} min")
```

Output:

```
stepped rod, 108 U: [41. 39. 30.]
   N    linear
   8  3.60e-03
  16  9.03e-04
  32  2.26e-04
  64  5.65e-05
linear, N = 8, largest nodal error: 1.54e-03
seiche, 80 elements: 0.000 2.000 6.002 12.011 20.042
n = 1 period: 1586 s = 26.4 min
```

S Supplementary Information

This section is reading. It is not part of the lecture.

S.1 A test function that is not admissible

Take Example 2.1 with $v = 1$, which does not vanish at $x = 1$. Now $v' = 0$, so the left side of (5) is 0, while $\int_0^1 6x\,dx = 3$. The difference is the boundary term of (4) at $x = 1$: $-u'(1)\,v(1) = -(-3)(1) = 3$. It is the force of the pin, which holds the total load 3.

S.2 The test function as a virtual displacement

In the rod, the weak form is a statement about work. On a slice of width dx the tension on the two faces gives the net force $w(x+dx) - w(x) = w'\,dx$, and equilibrium with the load $f\,dx$ means $w' + f = 0$. Now give the rod a small extra displacement $v(x)$, called virtual because nothing causes it. The load does the work $\int_0^1 f\,v\,dx$. Each slice stretches by a further $v'\,dx$ under the tension w, so the rod stores the work $\int_0^1 w\,v'\,dx$. Equilibrium says that these two are equal: multiply $w' + f = 0$ by v and integrate by parts as in Section 2. With $w = c\,u'$ this reads

$$\int_0^1 c\,u'\,v'\,dx = \int_0^1 f\,v\,dx,$$

which is (5). The rod is pinned at $x = 1$, so a virtual displacement cannot move that end, and $v(1) = 0$ is the admissibility condition again. With both ends pinned, $v(0) = v(1) = 0$.

S.3 The transpose from a difference matrix

Section 3 found $A^{\mathsf{T}} = -d/dx$ from the inner product. A difference matrix gives the same answer. Take the free–fixed grid of Section 4 with $N = 4$, unknowns $u_0, \dots, u_3$ and $u_4 = 0$. Forward differences $e_j = (u_{j+1} - u_j)/h$ (the $e_{j+1/2}$ of Section 1, and w_j likewise) give

$$\boldsymbol{A} = \frac{1}{h}\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}, \qquad \boldsymbol{A}^{\mathsf{T}} = \frac{1}{h}\begin{bmatrix} -1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

There is no fifth column in $\boldsymbol{A}$, because $u_4 = 0$ is the fixed end. A typical row of $\boldsymbol{A}^{\mathsf{T}}$ gives $(\boldsymbol{A}^{\mathsf{T}}\boldsymbol{w})_j = (w_{j-1} - w_j)/h = -(w_j - w_{j-1})/h$, a backward difference with a minus sign, which approximates $-dw/dx$. The first row is $-w_0/h$, the same formula with $w_{-1} = 0$: the free end. Multiplying the two matrices gives $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{A} = \boldsymbol{T}/h^2$.

S.4 Why the nodal values are exact

Let G_i be the Green's function for node i (Lecture 5): $-(c\,G_i')' = \delta(x - x_i)$ with the same boundary conditions. Test this equation against u, and test the equation for u against G_i:

$$u(x_i) = \int_0^1 c\,G_i'\,u'\,dx = \int_0^1 f\,G_i\,dx.$$

Away from x_i the quantity $c\,G_i'$ is constant. When c is constant on each element, G_i' is then constant on each element, so G_i is piecewise linear with kinks only at nodes. It lies in the trial

space: $G_i = \sum_j G_i(x_j)\,\phi_j$. Therefore

$$u(x_i) = \int_0^1 G_i\, f\, dx = \sum_j G_i(x_j) \int_0^1 \phi_j\, f\, dx = \sum_j G_i(x_j)\, F_j = U_i.$$

For the last step, test the equation for G_i against ϕ_j. This gives $\int_0^1 c\, G_i'\, \phi_j'\, dx = \phi_j(x_i)$, which is 1 for $j = i$ and 0 otherwise. Put $G_i = \sum_m G_i(x_m)\,\phi_m$ on the left: this says $\boldsymbol{K}\boldsymbol{g}_i = \boldsymbol{e}_i$, where $\boldsymbol{g}_i$ lists the nodal values of G_i. So $\sum_j G_i(x_j)\, F_j = \boldsymbol{g}_i^{\mathsf{T}}\boldsymbol{F} = \boldsymbol{e}_i^{\mathsf{T}}\boldsymbol{K}^{-1}\boldsymbol{F} = U_i$, since $\boldsymbol{K}^{-1}$ is symmetric.

For the stepped rod, node 1 gives $(G_1(x_0), G_1(x_1), G_1(x_2)) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{3})$, and $\boldsymbol{K}$ times this vector is $(0, 1, 0)$. The argument needs c constant on each element. When c varies inside the elements, as in Example 6.1, the nodal values are no longer exact.

S.5 More on quadratics

Example S.1 (One bubble)**. Given.** $-u'' = 4$ on $[0, 1]$ with $u(0) = u(1) = 0$. Use a single element and the single trial function $b(x) = 4x(1 - x)$; both end hats are dropped because both ends are fixed.

Find. (a) What are $K = \int_0^1 (b')^2\, dx$ and $F = \int_0^1 4b\, dx$? (b) What is the coefficient U, and how good is the approximation $U\, b(x)$?

Answer. (a) With $b' = 4 - 8x$,

$$K = \int_0^1 (4 - 8x)^2\, dx = \int_0^1 (16 - 64x + 64x^2)\, dx = 16 - 32 + \frac{64}{3} = \frac{16}{3},$$
$$F = \int_0^1 (16x - 16x^2)\, dx = 8 - \frac{16}{3} = \frac{8}{3}.$$

(b) $U = F/K = \frac{8}{3} \cdot \frac{3}{16} = \frac{1}{2}$, so $U\, b(x) = \frac{1}{2} \cdot 4x(1 - x) = 2x(1 - x)$. This satisfies $-u'' = 4$ and both boundary conditions: it is the exact solution. The exact solution was already in the trial space. Also $\frac{h^2}{8}\,|u''| = \frac{1}{2}$ is the coefficient of the bubble, as (8) predicts.

The formula (7) is unchanged, with i and j running over hats and bubbles. A bubble lives on one element, so it meets only that element's two hats and itself. In the order hat, bubble, hat every entry lies within two places of the diagonal, so the bandwidth is 2.

The hat–bubble entries exist only when c varies inside an element. The slope of a hat is the constant $\pm 1/h$ on the element, so

$$K_{\text{hat, bubble}} = \int_{x_{k-1}}^{x_k} c\, \phi'\, b_k'\, dx = \pm\frac{1}{h} \int_{x_{k-1}}^{x_k} c\, b_k'\, dx.$$

If c is constant on the element this is $\pm\frac{c}{h}\big[b_k\big]_{x_{k-1}}^{x_k} = 0$, because the bubble vanishes at both ends. For constant c, then, hats and bubbles decouple, and $\boldsymbol{K}$ is the hat matrix of Section 4 plus a diagonal for the bubbles. On element k put $x = x_{k-1} + sh$, so that $dx = h\, ds$ and $b_k' = (4 - 8s)/h$. Then $\int c\,(b_k')^2\, dx = \frac{c}{h}\int_0^1 (4 - 8s)^2\, ds = \frac{c}{h} \cdot \frac{16}{3}$, by the integral of Example S.1. The element matrix for (left hat, right hat, bubble) is therefore

$$\frac{c}{h}\begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & \frac{16}{3} \end{bmatrix}.$$

The nodal values do not change when bubbles are added. The gain is between the nodes.

If c varies smoothly inside the element, integrate by parts: $\int c\, b_k'\, dx = -\int c'\, b_k\, dx$, which is generally not zero. For $c = 1 + x$ we have $c' = 1$ and $\int b_k\, dx = h\int_0^1 4s(1-s)\, ds = \frac{2}{3}h$, so $K_{\text{hat, bubble}} = \pm\frac{1}{h}\cdot(-\frac{2}{3}h)$ on every element: $+\frac{2}{3}$ for the left hat and $-\frac{2}{3}$ for the right hat.

On the stepped rod c is constant on each element, so each bubble is found alone, with load $\frac{2}{3}h$ and diagonal entry $\frac{c_k}{h}\cdot\frac{16}{3}$:

$$B_k = \frac{2}{3}h\cdot\frac{3h}{16\,c_k} = \frac{h^2}{8c_k}.$$

This gives $\frac{1}{216}, \frac{1}{144}, \frac{1}{72}$, and $U(x)$ becomes the exact solution at every x. At the free end the slope is now $-\frac{1}{18} + \frac{1}{216}\cdot\frac{4}{h} = 0$.

More accuracy is not more smoothness

With or without bubbles, $U(x)$ is continuous, and nothing forces its slope to be continuous at the nodes: it is C^0. A bubble vanishes at both ends of its element but has slope $\pm 4/h$ there, so it does change the slopes at the nodes. They agree only if the computed coefficients happen to make them agree.

S.6 Cubic elements

A beam of length L and bending stiffness EI under load f obeys $(EI\,u'')'' = f$. Test against v and integrate by parts twice:

$$\begin{aligned}\int_0^L v\,(EI\,u'')''\,dx &= \Big[v\,(EI\,u'')'\Big]_0^L - \int_0^L v'\,(EI\,u'')'\,dx \\ &= \Big[v\,(EI\,u'')'\Big]_0^L - \Big[v'\,EI\,u''\Big]_0^L + \int_0^L v''\,EI\,u''\,dx.\end{aligned}$$

Assuming that the end conditions remove both brackets,

$$\int_0^L EI\,u''\,v''\,dx = \int_0^L f\,v\,dx.$$

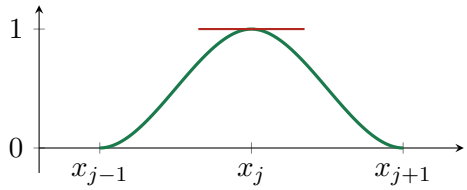
The unknowns at each node are a value and a slope; a paraxial ray carries the same two numbers, height and slope, at each plane. We expand

$$U(x) = \sum_j U_j\,\phi_j^d(x) + \sum_j U_j'\,\phi_j^s(x), \tag{10}$$

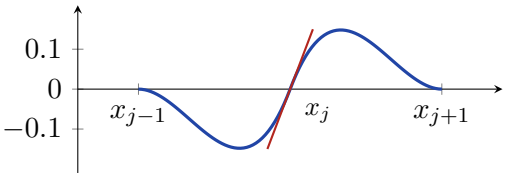
where U_j and U_j' are the value and the slope of U at node j. The displacement function ϕ_j^d and the slope function ϕ_j^s are cubic on each of the two elements touching node j and zero elsewhere. Both have value 0 and slope 0 at $x_{j\pm1}$. At x_j they are the two basis vectors of (value, slope):

$$\phi_j^d(x_j) = 1, \quad (\phi_j^d)'(x_j) = 0; \qquad \phi_j^s(x_j) = 0, \quad (\phi_j^s)'(x_j) = 1.$$

displacement function ϕ_j^d



slope function ϕ_j^s, with $h = 1$



The red segments mark the slope at x_j. Because U and U' are shared across each node, $U(x)$ is C^1.

Exercise S.2. Given. The element $[0,h]$ with $x_{j-1}=0$ and $x_j=h$, and a cubic $a_0+a_1x+a_2x^2+a_3x^3$.

Find. (a) Which cubic has $\phi_j^d(0)=0$, $(\phi_j^d)'(0)=0$, $\phi_j^d(h)=1$, $(\phi_j^d)'(h)=0$? (b) Which cubic has $\phi_j^s(0)=0$, $(\phi_j^s)'(0)=0$, $\phi_j^s(h)=0$, $(\phi_j^s)'(h)=1$? (c) Do your answers agree with the drawings?

Answer. In Section S.8.

S.7 Cubics on the rod, and the load of Example 6.1

The cubics also work for the rod, with the same formula (7). There are two unknowns per node, and in the order (value, slope, value, slope, ...) each row has at most six non-zeros, within three places of the diagonal: the bandwidth is 3. Only one number is removed at the fixed end, the value at $x=1$. The slope there stays unknown, and so does the slope at the free end, because $u'(0)=0$ is natural.

For Example 6.1 take $u=\cos(\pi x/2)$ and $c=1+x$, so that $c\,u'=-(1+x)\frac{\pi}{2}\sin\frac{\pi x}{2}$. The product rule gives

$$(c\,u')'=-\frac{\pi}{2}\sin\frac{\pi x}{2}-(1+x)\Big(\frac{\pi}{2}\Big)^2\cos\frac{\pi x}{2},$$
$$f=-(c\,u')'=\frac{\pi}{2}\sin\frac{\pi x}{2}+(1+x)\Big(\frac{\pi}{2}\Big)^2\cos\frac{\pi x}{2}.$$

Also $u'(0)=0$ and $u(1)=0$, as required.

S.8 Answer to Exercise S.2

On the element $[0,h]$ the four conditions on $a_0+a_1x+a_2x^2+a_3x^3$ are

$$\begin{bmatrix}1&0&0&0\\0&1&0&0\\1&h&h^2&h^3\\0&1&2h&3h^2\end{bmatrix}\begin{bmatrix}a_0\\a_1\\a_2\\a_3\end{bmatrix}=\begin{bmatrix}\text{value at }0\\\text{slope at }0\\\text{value at }h\\\text{slope at }h\end{bmatrix}.$$

For ϕ_j^d the right side is $(0,0,1,0)$. The first two rows give $a_0=a_1=0$, and the last two give

$$a_2h^2+a_3h^3=1,\qquad 2a_2h+3a_3h^2=0.$$

The second gives $a_2=-\frac{3}{2}a_3h$. Then the first gives $-\frac{1}{2}a_3h^3=1$, so $a_3=-2/h^3$ and $a_2=3/h^2$. With $s=(x-x_{j-1})/h$,

$$\phi_j^d=3s^2-2s^3.$$

For ϕ_j^s the right side is $(0,0,0,1)$. Again $a_0=a_1=0$, and

$$a_2h^2+a_3h^3=0,\qquad 2a_2h+3a_3h^2=1.$$

The first gives $a_2=-a_3h$. Then the second gives $a_3h^2=1$, so $a_3=1/h^2$ and $a_2=-1/h$. Therefore

$$\phi_j^s=h\,(s^3-s^2).$$

On the right element, with $t=(x-x_j)/h$, the same steps give $\phi_j^d=1-3t^2+2t^3$ and $\phi_j^s=h\,t(1-t)^2$. The largest size of ϕ_j^s on either element is $\frac{4}{27}h$, at distance $\frac{1}{3}h$ from x_j.